

Accurate Trajectory Recovery in Underserved Areas via Location Inference from Web Crowdsourced Data

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Abstract

In remote or underserved regions, where road networks are often either unavailable or poorly mapped, and GPS signals are sparse and unreliable, the quality of individual trajectories is severely compromised. In such contexts, web crowdsourced data becomes essential for accurately recovering trajectories and compensating for missing spatial information in the absence of explicit road networks. However, this situation introduces two key challenges: (i) scarcity of data in unseen regions, restricting transfer learning; and (ii) the necessity to infer latent movement structures under roadless conditions. To address these, we propose Region-aware Hierarchical Trajectory Recovery (RHTR) model, designed for location inference from web crowdsourced data in sparse, roadless scenarios. RHTR constructs multi-scale implicit grid-based offset maps from historical location data, with coarse grids capturing global patterns and fine grids refining local details. These sequential representations form region-aware encodings by sampling information around observed points, facilitating trajectory recovery. A coarse-to-fine mechanism leverages contextual information to progressively reconstruct missing segments. Experiments on two public datasets—simulating underserved and disaster-like settings with cross-region transfer—demonstrate that RHTR achieves state-of-the-art performance in trajectory recovery.

CCS Concepts

• Information systems → Spatial-temporal systems; Crowdsourcing.

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Keywords

Web Crowdsourced Data; Trajectory Recovery; Cross-regional modeling; Hierarchical learning.

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1 Introduction

In today's web and mobile ecosystems, applications increasingly rely on web crowdsourced mobility traces to model collective spatial perception and human movement patterns. Individual users typically produce sparse, irregular trajectories that cover only a tiny fraction of the environment; aggregating many such traces via web platforms yields rich spatial priors that in turn enable trajectory recovery, i.e., reconstructing travel paths from sparse GPS observations, especially in unmapped or roadless areas. This collective data powers critical functions like disaster mitigation, equitable mobility planning, community paths, and broad location-based services, all of which underscore the need for robust trajectory quality. Trajectory recovery has therefore drawn considerable research interest, with state-of-the-art methods delivering strong results in well-mapped urban settings featuring dense, up-to-date road networks [27]. Although road-based methods dominate the trajectory recovery field, they face significant limitations in areas where infrastructure is compromised, especially in road-network-free trajectory recovery for emergency response and disaster mitigation scenarios [19]. In these cases, individual trajectories become even sparser, communication infrastructure may be partially or totally degraded, and the road network is often sparsely covered, outdated, or completely unavailable. This makes road-network-dependent methods inapplicable, highlighting the importance of robust, crowdsourced, map-agnostic trajectory recovery. Models trained on dense,

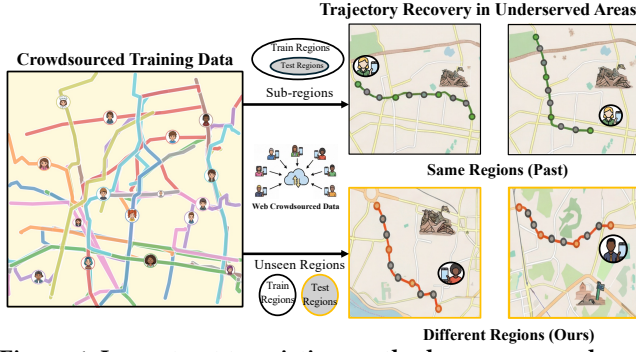


Figure 1: In contrast to existing methods, our approach enables the reconstruction of sparse trajectories in unknown domains by leveraging from web crowdsourced location data

high-quality urban data often underperform when applied to such domains [13]. As shown in Fig. 1, other methods are typically evaluated on varied trajectories in familiar regions, but in damaged areas, the environment may also change. Therefore, our focus is on trajectory recovery in previously unseen regions using web crowdsourced data. In this work, we directly address the dual challenges: trajectory recovery from sparse GPS in no-road-network scenarios, and cross-region generalization under severe cold-start conditions. We achieve this by harnessing low-cost, phone-collected, and web crowdsourced GPS data, enabling advanced recovery methods to be transferable and replicable in the world’s most underserved areas. A common strategy in recent trajectory recovery models is to discretize geographic space into a grid and encode each trajectory as a sequence of grid cells, analogous to word embeddings in natural language processing [2, 36]. Grid-based modeling offers clear advantages: it regularizes continuous coordinates into a compact vocabulary, facilitates the use of sequence models over cell indices, and provides a natural way to incorporate local spatial context. At the same time, it introduces new dependencies and failure modes. The quality of the recovered trajectories becomes highly sensitive to the grid scale: overly coarse grids smear out fine-grained motion patterns, while overly fine grids fragment trajectories and exacerbate data sparsity. Incorporating grid connectivity (e.g., adjacency or higher-order neighborhood relationships) can partially mitigate these issues [1]. Consequently, how to design grid-based representations that remain efficient, robust, and transferable across regions with dramatically different densities and map conditions emerges as a central research problem and is one that our proposed model is specifically designed to address. Our approach leverages the flexibility of grid partitioning to learn generalizable spatial representations, even under severely sparse sampling. At coarse, middle, and fine scales, each grid cell is represented by a learned embedding that aggregates local trajectory evidence from sparse GPS points together with relational information from neighboring cells via a deformable grid encoder and neighbor sampling. This yields transferable spatial embeddings that remain robust in unseen areas and under distribution shift. To address the multi-scale nature of geographical data, we employ a Hierarchical Multi-Scale Grid Transformer that propagates information across scales with an attention-based method, enforcing consistency between fine- and

coarse-level recoveries. Together, these components directly tackle the dual challenges of cross-region generalization and multi-scale spatial reasoning, as well as trajectory recovery from sparse GPS traces, as validated by extensive experiments on multiple real-world datasets. Our contributions are as follows:

- We propose a grid-based formulation of road-network-free, cross-region trajectory recovery from *sparse* GPS observations, where irregular, low-frequency traces are represented on nested coarse–middle–fine grids, enabling region-agnostic spatial priors that transfer to unseen areas.
- We design **RHTR**, a Region-aware Hierarchical Trajectory Recovery model that couples a deformable multi-scale grid encoder with GPS grid fusion, cross-scale neighborhood sampling, and fine-scale offset regression, explicitly enforcing consistency between coarse skeletons and fine trajectory details while mitigating spatial drift caused by large sampling gaps.
- We conduct extensive experiments on *Geolife* and *Porto*, including enlarged ROIs and severe sparsity regimes, showing that **RHTR** achieves state-of-the-art results on both classification and spatial metrics. Ablation studies further confirm the complementary roles of the grid map, multi-scale hierarchy, neighbor sampling, and deformable alignment in handling sparsity and distribution shift.

2 Related Work

2.1 Network-free Trajectory Recovery

Network-free trajectory recovery techniques aim to reconstruct trajectories without relying on predefined road networks. Early studies [4, 7, 12, 24] explore forecasting the chain of visited spots linking start and end points, employing tactics such as optimization frameworks, established reference markers for adjustment, or Markov-based techniques. With the advent of deep learning, especially with the development of time series models [34, 35], sequential modeling paradigms leveraging RNNs [8, 15, 29] and Transformers [28] have been increasingly adopted to tackle trajectory-related challenges, while Si et al. [23] propose another transformer-based method for recovering implicit trajectories from sparse data. Xia et al. [32] propose a novel attentional-based framework to address the problem of data sparsity and to take long-term history into account. Their main contributions come in the form of measured use of historical data to augment the learning of trajectories. Sun et al. [25] propose another architecture that combines measured model building with analysis of temporally varying human movement patterns, which could include both full and sparse movement data. Their model is mainly based on the use of graph neural networks (GNN) which are able to capture temporal patterns and complex location transition information. Elsharif et al. [10] employ a heuristic algorithm to enable long-distance trajectory completion. However, the aforementioned methods are typically evaluated within the same regions as their training data, lacking geographic awareness capabilities. In contrast, our model leverages web crowdsourced location data to learn spatial variations in routes, thereby capturing the intrinsic logic underlying true trajectory dynamics.

2.2 Grid-based Trajectory Recovery

Grid-based methods focus on simplifying road network or trajectory representation learning by discretizing spatial domains into individual cells [18, 22, 33]. As a pioneering effort in the field, Li et al. [18] adopt an encoder-decoder structure [26] to derive durable grid-aligned trajectory embeddings from sparse datasets. In particular, they do not rely on the presence of pre-determined road maps, which alleviates the need for additional map-matching procedures. Chang et al. [3] proposes a method for trajectory recovery based on integration of contrastive learning with an attention-based backbone encoder. Their model introduces a novel similarity measure that alleviates some known problems with conventional metrics, such as sensitivity to measurement errors and inefficiency. Chen et al. [6] propose a framework that combines road network (via grid discretization) with GPS trajectory representation learning for trajectory recovery. Their main module is the GridGNN, which learns spatial features for each road segment in their respective grids. Finally, Zhou et al. [36] leverage the complementary strengths of grid-based spatial discretization for capturing regional patterns, road network topology for structural semantics, and raw GPS sequences for fine-grained spatio-temporal details, resulting in enhanced trajectory representation. Although grid-based discretization approaches have substantially augmented the representational capacity of trajectories, prevailing methods often underutilize the inherent advantages of grid encoding. In particular, distinct from alternative techniques, our methodology capitalizes on the positional encoding embedded within grid maps, which inherently captures the distinctive spatial interrelationships among grid cells.

2.3 Cross-region Generalizability for Trajectory Recovery

Cross-region generalizability refers to the ability of our learned model to produce useful inference results outside of its learned domain. For the trajectory recovery task, cross-region generalizability has not been fully explored. Most such methods [11, 16, 20] are based on the learning of trajectory encoders that map physical trajectories to embedding vectors. With the advent of large-scale models, large language models (LLMs) demonstrate inherent generalization prowess [14, 21], thereby catalyzing advancements in research on the generalizability of trajectory representations. For example, Wei et al. [31] harness LLMs to interpret road networks alongside peripheral points of interest (POI) data, thereby augmenting the representational efficacy of trajectories, while incorporating a multi-expert mechanism to discern intricate motion states within trajectories. Likewise, Zhou et al. [37] exploit LLMs to assimilate trajectory motion patterns and underlying travel intents, facilitating high-fidelity performance in downstream trajectory-centric applications. Nevertheless, prevailing approaches have yet to leverage web crowdsourced location data for deriving spatial embeddings of mobility patterns, which could bolster trajectory generalization across unfamiliar domains. This forms the pivotal core of our method: by inferring precise locations via web crowdsourced location data, we enable robust and accurate trajectory recovery tasks.

3 Proposed Approach

In this section, we introduce our Region-aware Hierarchical Trajectory Recovery (RHTR) model, designed for recovering dense user trajectories from sparse location data by leveraging multi-scale implicit spatial grid representations. Fig. 2 provides an overview of the RHTR model. The RHTR model comprises four key components: (1) multi-scale spatial grid map representation learning from web crowdsourced data, (2) trajectory representation learning, (3) multi-scale neighborhood sampling, and (4) hierarchical cross-attention for location inference. We go into each module in detail in the following.

3.1 Multi-scale Spatial Grid Map Representation Learning

We aggregate sparse user trajectories across the target region and discretize them into a uniform $G \times G$ grid. Cell-wise visitation densities are normalized to form a preliminary density-aware grid map. We then reframe this grid map as a semantically enriched multi-channel feature map $\mathbf{M} \in \mathbb{R}^{3 \times G \times G}$, where the first channel encodes the normalized visitation intensity for each cell g_i , while the remaining two channels embed the normalized centroid coordinates to ensure spatial awareness.

To further distill multi-scale adaptive representations from the crowdsourced grid map, we introduce a hierarchical spatial map extraction method, inspired by Deformable Convolutional Networks[9]. This multi-scale approach enables the model to capture spatial hierarchies at varying scales, accommodating diverse urban structures such as fine-grained local patterns in dense areas and broader regional contexts in sparse zones. By employing deformable convolutions with decreasing kernel sizes and offset scales across stages, the network dynamically adjusts receptive fields to irregular mobility distributions, enhancing robustness to variations in crowdsourced data density and geometry. Specifically, the input feature map $\mathbf{M} \in \mathbb{R}^{3 \times G \times G}$ undergoes an initial convolutional block to project it into a higher-dimensional space:

$$\mathbf{X}_1 = \text{ConvBA}(\mathbf{M}), \quad (1)$$

where $\mathbf{X}_1 \in \mathbb{R}^{D \times G \times G}$ and D is the hidden embedding dimension. We then apply a ConvBA block—a composite layer consisting of a convolutional neural network (CNN) [17] followed by batch normalization (BatchNorm) and ReLU activation. This is followed by a series of deformable stages interleaved with downsampling operations. For each stage l (where $l = 1, \dots, L$ and L is the number of stages), the process is:

$$\mathbf{O}_l, \text{offset}_l = \text{DeformStage}(\mathbf{X}_l; k_l, \sigma_l), \quad (2)$$

$$\mathbf{X}_{l+1} = \text{ConvBA}(\mathbf{O}_l + \mathbf{X}_l; s = 2), \quad (3)$$

where k_l denotes the kernel size and σ_l is the offset scale for stage l , with k_l and σ_l decreasing to focus on finer details at deeper levels. Within each deformable stage, we first predict adaptive offsets to enable flexible deformation of the convolutional kernels, allowing the network to focus on relevant spatial features in the crowdsourced grid. Given the input $\mathbf{X}_l \in \mathbb{R}^{D \times H_l \times W_l}$, the offsets are predicted using a standard convolution:

$$\text{offset}_l = \tanh(\text{Conv}_{k_l \times k_l}(\mathbf{X}_l)) \cdot \sigma_l, \quad (4)$$

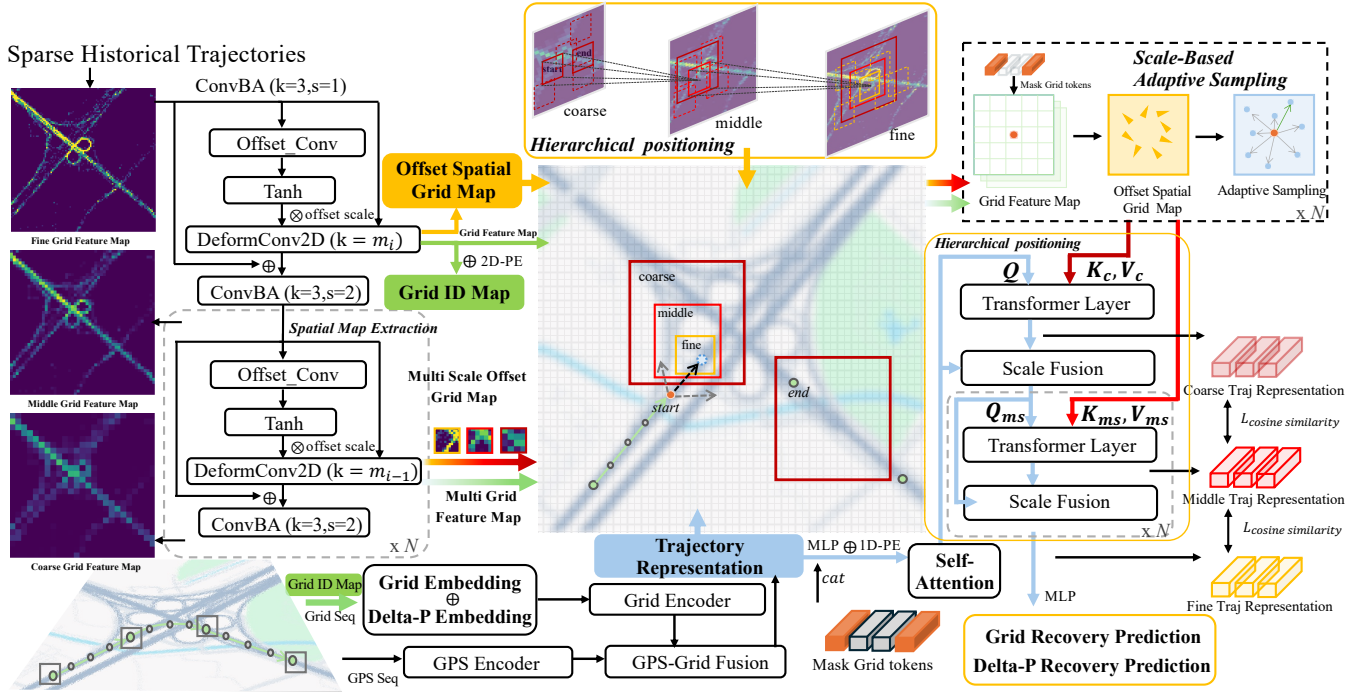


Figure 2: Overview of the RHTR model

where $\text{Conv}_{k_l \times k_l}$ is a $k_l \times k_l$ convolutional layer outputting $2k_l^2$ channels (corresponding to x and y offsets for each of the k_l^2 kernel positions), $\tanh(\cdot)$ constrains the offset range to $[-1, 1]$, and σ_l is the offset scale that modulates the deformation extent. The offsets are initialized to zero, ensuring the stage initially behaves like a regular convolution. Next, the deformable convolution aggregates features using these offsets:

$$\tilde{O}_l = \text{DeformConv}_{k_l \times k_l}(\mathbf{X}_l, \text{offset}_l), \quad (5)$$

where $\text{DeformConv}_{k_l \times k_l}$ is a deformable convolutional layer with kernel size k_l and stride 1 to maintain spatial dimensions. This is followed by batch normalization and ReLU activation to stabilize and non-linearize the features:

$$\mathbf{O}_l = \text{ReLU}(\text{BatchNorm}(\tilde{O}_l)), \quad (6)$$

this design allows each stage to adaptively adjust its sampling locations based on the input, effectively capturing non-rigid deformations and varying densities in the mobility patterns, while preserving spatial dimensions in the deformable convolution and halving them via strided downsampling. To incorporate positional information and prepare for subsequent trajectory inference, we augment each stage's output feature map $\mathbf{O}_l$ with 2D sinusoidal positional encodings:

$$\mathbf{O}'_l = \mathbf{O}_l + \text{PE}_l, \quad (7)$$

the augmented maps are then reshaped and flipped vertically to yield flattened cell embeddings $\{\mathbf{E}_l\}_{l=1}^L$, which serve as the ID representations for the cells in the grid map at each scale level, facilitating

bottom-up indexing aligned with trajectory sequences and providing spatial information for obtaining sequential representations of cell IDs in downstream processing.

3.2 Trajectory Representation Learning

The trajectory representation learning module is designed to encode sparse GPS trajectories into contextualized vectors that capture both spatial and movement dynamics, facilitating accurate inference in subsequent hierarchical modules. By integrating fine-grained grid embeddings with GPS coordinates and their respective offsets (deltas, see below), this module learns robust representations resilient to sparsity and noise in real-world data. The multi-scale representations $\{\mathbf{E}_l\}_{l=1}^L$ form the embedding table for grid cell IDs. We start with the finest-grained grid embeddings $\mathbf{E}_1$. Now given a sparse GPS trajectory $T = \langle p_1, p_2, \dots, p_m \rangle$ with $p_i = (x_i, y_i, t_i)$, we map it to a grid sequence $T_g = \langle g_1, g_2, \dots, g_m \rangle$ and compute the delta sequence $\Delta T = \langle \delta_1, \delta_2, \dots, \delta_m \rangle$, where $\delta_i = (x_i - x_{c_i}, y_i - y_{c_i})$, with (x_{c_i}, y_{c_i}) being the bottom-left coordinates of the cell containing p_i . To support masking during inference, we append a learnable mask token to $\mathbf{E}_1$, yielding an extended table $\mathbf{E}_1^f \in \mathbb{R}^{(G \times G + 1) \times D}$, where $G \times G$ is the number of fine-grained cells. The grid sequence embeddings are obtained by looking up T_g in $\mathbf{E}_1^f$. To obtain a more robust trajectory representation that mitigates the limitations of sparse data, such as incomplete sampling or noise in GPS signals, while enhancing the ability to model long-range dependencies and mobility patterns, we first compute the grid sequence representation by projecting the delta sequence using a multi-layer perceptron (MLP) consisting of two linear layers with ReLU activation. The projected deltas are then added to the grid sequence embeddings,

and the fused sequence is processed through an LSTM encoder, followed by mean pooling. In parallel, the GPS sequence representation is derived by projecting the raw GPS sequence to the embedding dimension, feeding it into another LSTM to get GPS sequence representation. These two representations are concatenated and subsequently mapped through an MLP to produce a comprehensive trajectory embedding $\mathbf{e}_s$. This design leverages both the GPS sequence, which captures continuous movement dynamics and fine-grained positional shifts, and the grid sequence, which provides discrete spatial context and regional semantics.

3.3 Multi-scale Neighborhood Sampling

To support hierarchical cross-attention for location inference, this module adaptively samples neighborhoods around each trajectory point on the multi-scale grid representations, providing scale-specific key-value (KV) pairs that encode localized spatial contexts at varying scales for progressive location refinement. Given a masked grid sequence T_g of length m at the fine scale ($S_1 = G$), denoted as $T_g^{(1)} = \langle g_1^{(1)}, g_2^{(1)}, \dots, g_m^{(1)} \rangle$, we remap it to coarser scales. Due to the convolution operations, each layer reduces the scale by half as l increases. Then, we obtain the spatial positions of the trajectory at different scales, denoted as $T_g^{(l)}$. At each scale l , for every trajectory point, a $K_l \times K_l$ neighborhood is sampled around $g_i^{(l)}$ using the deformable feature map $\mathbf{O}_l \in \mathbb{R}^{D \times S_l \times S_l}$ and offsets $\mathbf{offset}_l \in \mathbb{R}^{2K_l^2 \times S_l \times S_l}$. To perform this Scale-Based Adaptive Sampling, we first convert the grid index $g_i^{(l)}$ into spatial coordinates $(r_i^{(l)}, c_i^{(l)})$ on the feature map, aligning with its layout to ensure accurate positioning. A base sampling grid is then constructed as a symmetric kernel centered at $(r_i^{(l)}, c_i^{(l)})$, where for each $y, x \in \{0, \dots, K_l - 1\}$, the offsets are computed as $dy^k = (y - \lfloor K_l/2 \rfloor) \cdot \alpha$ and $dx^k = (x - \lfloor K_l/2 \rfloor) \cdot \alpha$, with α being the dilation factor to expand the receptive field without increasing kernel size, allowing for broader context capture at coarser scales. Learned offsets, predicted per location and corresponding to each of the $K_l \times K_l$ kernel positions, are gathered from $\mathbf{offset}_l$ at $(r_i^{(l)}, c_i^{(l)})$, yielding $\Delta \in \mathbb{R}^{2 \times K_l^2}$ with entries (Δ_y^k, Δ_x^k) for the k -th kernel position. This deformable mechanism, building on deformable convolutional networks [9], introduces flexibility by allowing non-uniform sampling, which is crucial for handling irregular trajectory patterns. The final sampling positions are computed as:

$$\mathbf{P}^k = (r_i^{(l)} + dy^k + \Delta_y^k, c_i^{(l)} + dx^k + \Delta_x^k), \quad \forall k = 1, \dots, K_l^2, \quad (8)$$

where (dy^k, dx^k) are from the base grid. These positions enable adaptive focus on salient features, such as nearby landmarks or road segments, enhancing location refinement. To facilitate interpolation, the positions are normalized to the grid sampling range $[-1, 1]$. Bilinear interpolation is then applied on the feature map $\mathbf{O}_l$ using the normalized grid, resulting in sampled features $\mathbf{N}_l \in \mathbb{R}^{D \times m \times K_l^2}$, which are permuted to $[K_l^2, m, D]$ for cross-attention. This output serves as the KV pairs, where the aggregation across K_l^2 neighbors provides a rich, localized embedding. By progressively refining from coarse to fine scales, the hierarchical approach reduces ambiguity in masked location inference, as coarser scales offer global context while finer ones pinpoint details.

3.4 Hierarchical Cross-attention for Location Inference

To enable progressive refinement of masked trajectory points through hierarchical positioning, we employ a hierarchical cross-attention mechanism that integrates multi-scale neighborhood features, allowing the model to iteratively narrow down location predictions from coarse to fine scales. This approach leverages the adaptive KV pairs from the multi-scale sampling module ($\mathbf{N}_l$ for each scale l), where coarser scales provide broad contextual guidance to reduce search space ambiguity, while finer scales enable precise localization. We initialize the query as $\mathbf{Q}^{(l+1)} = \mathbf{e}_s$ (the trajectory embedding, with masked positions using the learnable mask token). The refinement proceeds from coarse to fine scales, with l decrementing. At each stage, we perform a cross-attention operation following the standard Transformer formulation: the current-scale query attends to the neighboring representation from the previous (coarser) scale to produce an attention-based feature $\mathbf{F}^{(l)}$. The updated query for the next (finer) scale is then obtained via an MLP fusion:

$$\mathbf{Q}^{(l)} = \text{MLP}^{(l)}(\mathbf{Q}^{(l+1)} \parallel \mathbf{F}^{(l)}), \quad (9)$$

with $\parallel$ denoting concatenation along the feature dimension. Repeating this procedure yields trajectory representations across scales, producing $\mathbf{F}^{(l)}$ from coarse to fine. For prediction, we apply scale-specific classification heads to generate logits over the grid cells at each scale:

$$\mathbf{L}^{(l)} = \text{Head}^{(l)}(\mathbf{F}^{(l)}), \quad (10)$$

where each head is a linear projection followed by softmax normalization. Additionally, a regression head predicts fine-scale offsets within the predicted cell:

$$\Delta^{(1)} = \text{Head}_\delta(\mathbf{F}^{(1)}) \in \mathbb{R}^{m \times 2}, \quad (11)$$

representing normalized deltas (δ_x, δ_y) relative to the cell's bottom-left corner. Finally, the model outputs the predicted multi-scale grid IDs for the masked positions along with the fine-scale offsets. During training, we optimize the model using a composite loss that enforces accuracy across scales and consistency between hierarchical features. Ground-truth grid IDs at coarser scales $\mathbf{id}^{(l)}$ are derived from fine-scale labels $\mathbf{id}^{(1)}$. Multi-scale cross-entropy losses ensure correct cell classification at each scale:

$$\mathcal{L}_{ce} = \sum_{l=1}^L \text{CE}(\mathbf{L}^{(l)}, \mathbf{id}^{(l)}), \quad (12)$$

A GPS loss refines fine-scale predictions by reconstructing coordinates from the predicted fine-grid ID and offsets, using mean absolute error (MAE) against raw GPS coordinates in the batch:

$$\mathcal{L}_{gps} = \text{MAE}(\hat{\mathbf{p}}^{(1)}, \mathbf{p}_{gps}), \quad (13)$$

where $\hat{\mathbf{p}}^{(1)}$ is the predicted position derived from the coordinate corresponding to $\arg \max(\mathbf{L}^{(1)})$ plus $\Delta^{(1)}$, and $\mathbf{p}_{gps}$ is the normalized ground-truth GPS. To promote hierarchical consistency and ensure spatial similarity in trajectory representations across scales, we incorporate multi-scale KL divergence losses between consecutive feature pairs:

$$\mathcal{L}_{kl} = \sum_{l=2}^L \text{KL}(\sigma(\mathbf{F}^{(l)}), \sigma(\mathbf{F}^{(l-1)})), \quad (14)$$

where σ denotes the softmax function. The total loss is the sum of these three terms:

$$\mathcal{L} = \mathcal{L}_{ce} + \mathcal{L}_{gps} + \mathcal{L}_{kl}. \quad (15)$$

The hierarchical localization strategy in our model offers several key advantages, including enhanced inference efficiency through progressive refinement of the search space, superior accuracy in reconstructing sparse trajectories, and better handling of multi-scale spatial contexts compared to conventional flat attention mechanisms.

4 Experiments

4.1 Experimental Settings

4.1.1 Datasets. Our experiments are based on real-world trajectory datasets recorded from the web: the GeoLife dataset and the Porto dataset. These datasets capture the actual travel trajectories of users in the Beijing and Porto regions, respectively. For the GeoLife dataset, we first resample the original GPS trajectories and retain those with an effective sampling interval between 3–6 seconds, ensuring the quality of trajectory positions, while for Porto we sample with a 15-second interval, emphasizing a more sparse data scenario. In terms of spatial processing, we restrict all trajectories in both datasets to a predefined study area and divide this area into regular subregions, where each subregion contains the trajectories of multiple users within that area. Specifically, we divide the densely covered Beijing region in the GeoLife dataset into 16×16 subregions, while for the Porto dataset, it is divided into 14×14 subregions, with each subregion covering approximately 1km^2 . Additionally, we filter the position points based on the grid boundaries to align with the Underserved Areas scenario. After completing the above preprocessing, we partition the datasets based on the number of subregions, following an 8:1:1 ratio to create training, validation, and test sets. All experimental results are reported on the independent test set, with the validation set being used exclusively for model selection and hyperparameter tuning.

4.1.2 Baseline Methods. We benchmark our approach against a representative suite of trajectory-recovery baselines. TransformerID centers on the Transformer [28] architecture, fusing GPS data with grid IDs to learn trajectory representations. TERI [5] is a transformer-based two-stage framework for irregularly sampled traces, combining a detection stage with a recovery stage and a robust temporal encoding module. TrajBERT [23] treats trajectories as token sequences and leverages bidirectional transformer masking to recover missing segments from sparse observations. MMSTGED [30] integrates micro- and macro-level trajectory semantics on a learned graph, coupling node/edge representations with sequence attention for reconstruction. GREEN [36] is a hybrid method that fuses grid discretization, GPS observations, and road-network cues to enforce spatial consistency. AttenMove [32] employs attentional mechanisms over historical mobility context to mitigate sparsity and capture long-range dependencies, while PeriodicMove [25] augments representation learning with periodic/temporal regularities using graph-based encoders.

4.1.3 Evaluation Metrics. We evaluate trajectory recovery from two complementary perspectives: point-wise localization and overall trajectory quality. For localization, we report Accuracy, Precision,

Recall, and F_1 on a 100×100 grid, where Accuracy is the proportion of recovered points that fall into the correct cell, Precision and Recall measure false positives and false negatives respectively, and F_1 is their harmonic mean. For overall trajectory quality, we use Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Fréchet Distance (FD) between recovered and ground-truth trajectories, with FD capturing shape similarity as the minimal “walking distance” required to traverse both curves in synchrony. Together, these metrics jointly assess local localization accuracy and global trajectory fidelity in sparse, underserved regions.

4.1.4 Parameter Settings. We implement our model in PyTorch and train for up to 200 epochs with early stopping, using a batch size of 256, an initial learning rate of 1×10^{-3} , the AdamW optimizer, and a learning-rate scheduler with patience 5 and decay factor 0.5. The model hidden dimension is set to $d = 512$. The offset scales σ_l and kernel sizes k_l are set to 4, 2, 1 and 7, 5, 3 respectively for $l = 1, 2, 3$, respectively, enabling multi-scale receptive fields from coarse to fine for sparse GPS signals.

4.2 Performance Analysis

Table 1 reports the main comparison between RHTR and all baselines under three sampling regimes, $\mu \in \{4, 3, 2\}$ multiples of the dataset’s unit time interval, on both *Geolife* and *Porto*. Overall, RHTR achieves state-of-the-art results across conventional classification metrics (Accuracy, Precision, Recall, F_1) and localization metrics (MAE, FD), and maintains these gains as trajectories become sparser.

Robustness under sparse sampling. When observations are most sparse ($\mu = 4$), RHTR shows the clearest advantage. On *Geolife*, it outperforms the strongest baseline MMSTGED on both Acc and F_1 , while also reducing spatial error, as evidenced by the relative reduction in MAE. On *Porto* at $\mu = 4$, the margins are similar, with concurrent reduction in MAE and FD. On the other hand, we see that in terms of precision, the MMSTGED outperforms our model on the Porto dataset. MMSTGED’s precision advantage likely stems from its fixed interaction graph, which is well aligned with this dense, taxi-style road network and thus favors only high-confidence, high-traffic cells, reducing false positives. In contrast, RHTR deliberately explores a broader set of plausible cells via multi-scale deformable modeling and offset regression, slightly lowering precision but ensuring consistently high performance on other metrics. Our numbers indicate that multi-scale spatial reasoning, GPS-grid fusion, and offset-aware alignment extract more usable signal from low-frequency traces than attention-only or single-scale designs.

Enlarged ROI and map-level robustness. Table 3 further evaluates RHTR and baselines on an enlarged region of interest (ROI, $2 \times$ area), again at multiple sampling rates. The larger ROI introduces more diverse GPS configurations and increases the difficulty of correctly reconstructing underlying road structures, leading to performance drops for all models, especially in MAE and FD. Nonetheless, RHTR consistently attains the lowest FD as well. On *Porto*, the gains are more pronounced, relative to the best baseline across the same sampling regimes. These results indicate that, even when the spatial search space is doubled and the number of plausible configurations grows, RHTR’s multi-scale grid modeling and

Table 1: Performance comparison on Geolife and Porto datasets under different sampling intervals.

IS	Methods	Geolife						Porto					
		Acc(%)	Pre(%)	Recall(%)	F1(%)	MAE(m)	FD(m)	Acc(%)	Pre(%)	Recall(%)	F1(%)	MAE(m)	FD(m)
$\mu = 4\times$ time interval	TransformerID	20.7	12.36	15.75	11.32	16.87	27.56	4.5	4.87	4.45	3.08	49.71	77.12
	MMSTGED	37.29	28.36	26.66	24.83	13.1	25.02	30.51	38.92	23.45	25.16	55.81	96.51
	GREEN	27.6	18.26	19.09	16.1	16.63	29.13	14.53	18.74	14.5	13.69	47.61	78.99
	PeriodicMove	29.22	17.26	17.28	15.67	36.16	79.46	26.5	35.39	26.5	28.17	53.3	91.58
	TrajBERT	26.47	14.38	14.58	13.2	51.08	117.46	29.52	37.01	29.52	30.2	56.09	99.69
	TERI	28.9	16.3	16.49	14.81	22.84	41.05	27.33	33.68	27.31	27.44	48.25	78.09
	AttnMove	28.47	16.89	17.34	15.26	27.03	55.67	29.57	37.93	29.53	30.39	55.64	93.16
	Ours	43.57	31.7	31.57	29.49	12.07	20.82	30.53	38.09	30.5	31.06	39.68	66.77
$\mu = 3\times$ time interval	TransformerID	32.67	22.96	24.84	20.69	12.29	20.29	5.88	7.05	5.84	4.4	38.89	64.74
	MMSTGED	47.08	37.07	34.76	33.05	10.85	20.86	33.67	44.8	27.11	28.42	42.51	77.84
	GREEN	34.66	24.04	25.13	21.70	13.04	22.36	20.3	25.83	20.31	19.8	37.78	68.56
	PeriodicMove	32.09	20.34	20.48	18.30	26.01	58.64	29.35	38.37	29.35	31.18	43.81	80.79
	TrajBERT	32.61	19.20	19.33	17.77	42.52	106.15	33.23	41.11	33.13	33.99	44.63	84.14
	TERI	36.32	22.44	22.58	20.68	17.18	31.91	31.82	39.35	31.78	32.3	38.08	67.67
	AttnMove	37.52	24.51	24.94	22.59	19.60	43.89	32.91	41.75	32.85	33.91	44.93	80.66
	Ours	52.09	39.44	39.37	37.15	10.07	18.41	34.21	42.32	34.17	35.19	31.31	57.76
$\mu = 2\times$ time interval	TransformerID	32.93	22.99	25.85	21.12	11.83	19.59	8.91	11.11	3.85	8.9	27.83	48.57
	MMSTGED	55.9	46.85	43.8	42.2	9.71	17.71	35.33	52.54	34.98	36.45	31.45	60.52
	GREEN	43.84	31.67	32.11	29.11	11.21	19.62	25.87	32.97	25.9	25.78	26.65	51.96
	PeriodicMove	36.95	23.92	23.89	21.77	35.75	87.52	34.37	44.7	34.29	35.33	41.22	86.48
	TrajBERT	37.83	23.74	23.82	22.14	34.15	80.61	37.14	45.8	37.06	38.12	39.03	72.17
	TERI	44.13	30.02	29.97	27.94	14.38	27.22	36.84	45.22	36.77	37.51	27.73	52.90
	AttnMove	44.12	30.51	31.00	28.40	18.30	41.28	36.18	45.43	36.11	36.98	35.76	70.13
	Ours	61.31	48.53	48.34	46.16	9.61	19.11	38.23	48.34	38.13	39.35	23.24	46.46

deformable alignment preserve its advantage in both local accuracy and global trajectory shape.

Cross-region consistency and metric behavior. As sampling densifies, all methods predictably improve, but **RHTR**'s advantage persists for both datasets: it achieves the best performance on most metrics across both sampling intervals and datasets, hinting at its ability to learn from stronger spatial fidelities under differing mobility and topology distributions. Improvements are balanced across classification metrics (Acc/Pre/Recall/ F_1), with precision and recall rising together rather than trading off, while localization metrics (MAE/FD) see the largest relative gains in the sparsest regime. This behavior aligns with the design of multi-scale offset modeling and deformable sampling, which directly mitigate spatial drift accumulated between distant observations.

Baseline-specific observations. Among baselines, MMSTGED is often strongest at higher sampling densities due to its fusion of micro/macro semantics on a learned graph, yet **RHTR** consistently surpasses it as observations thin out, suggesting benefits beyond fixed-graph semantics. TERI, designed for irregular intervals, remains robust to non-uniform sampling but falls behind MMSTGED and **RHTR** in the highest-sparsity regimes, indicating that temporal encoding alone is insufficient without explicit multi-scale spatial fusion. TrajBERT is competitive on sequence-level metrics but lags on MAE and Fréchet Distance, especially under sparse

sampling, showing that masked-token pretraining cannot fully correct spatial misalignment without explicit spatial cues. GREEN and other grid-style methods benefit from discretization but suffer from cell-boundary artifacts and scale sensitivity; by contrast, **RHTR**'s deformable alignment alleviates these issues and yields lower MAE/FD at matched recall. Finally, AttenMove, PeriodicMove, and vanilla Transformer baselines leverage temporal attention and long-range context but still exhibit the largest deficits in spatial metrics, underscoring that temporal priors alone cannot replace explicit spatial offset modeling and cross-scale fusion. We summarize this section via Fig. 3, which qualitatively compares recovered trajectories from all baselines against the ground truth. It is clear that almost every baseline model introduce noticeable zig-zags, local kinks, or systematic shifts away from the true curved trajectory. In contrast, MMSTGED and especially **RHTR** produce trajectories that adhere much more closely to the ground-truth geometry, with **RHTR** best preserving both the global curvature and local alignment of points.

4.3 Ablation Study

In our ablation experiments we studied the influence of important modules on overall model performance. The largest relative performance drop occurs when the grid map is excluded. This is because the grid partitions geographical information into discrete

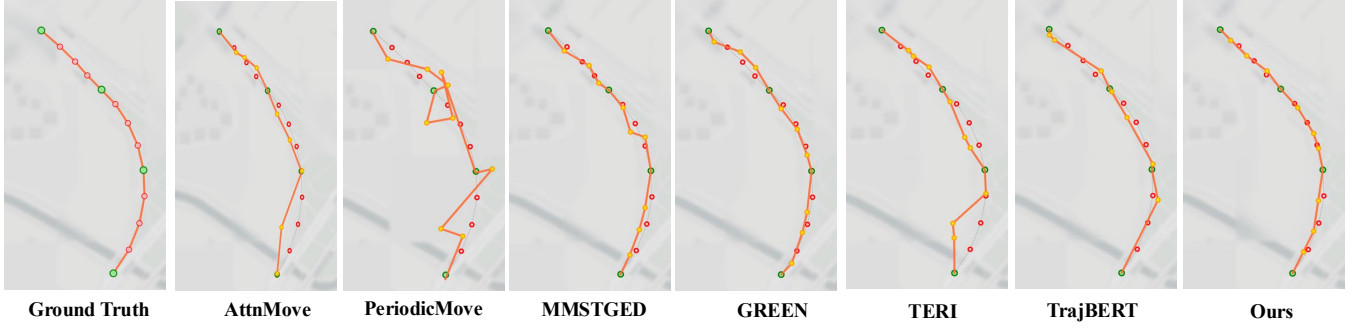


Figure 3: Performance comparison of different trajectory recovery models on sparse GPS data. Green points represent known observed trajectory points, red points indicate missing trajectory models to be recovered, and orange lines/points denote the recovery results from various models (including baselines and our RHTR method).

Table 2: Model Performance in a Large Area

SI	Model	Geolife			Porto		
		MAE	RMSE	FD	MAE	RMSE	FD
$\mu = 4 \times$ time interval	TransformerID	30.98	37.59	43.77	63.48	101.02	91.86
	MMSTGED	21.19	35.22	33.83	66.38	99.93	102.72
	GREEN	21.52	31.96	32.83	58.45	100.66	89.65
	PeriodicMove	31.88	83.7	72.24	89.88	206.55	171.38
	TrajBERT	36.05	88.93	79.67	118.27	323.39	280.06
	TERI	40.16	65.91	71.63	61.83	115.22	99.85
	AttnMove	29.4	99.05	68.02	75.84	132.3	127.13
	Ours	19.21	29.1	31.18	53.37	99.86	87.84
$\mu = 3 \times$ time interval	TransformerID	23.27	27.84	33.42	50.54	77.41	75.23
	MMSTGED	21.19	35.22	33.83	53.39	77.08	89.43
	GREEN	18.83	28.64	29.51	46.61	77.15	74.52
	PeriodicMove	25.78	60.32	58.94	114.27	267.34	201.65
	TrajBERT	37.17	128.56	107.49	289.04	99.52	241.02
	TERI	33.1	51.51	49.44	52.57	92.81	109.26
	AttnMove	26.04	96.26	61.05	62.08	108.28	107.03
	Ours	16.46	25.37	26.57	39.74	76.49	70.98
$\mu = 2 \times$ time interval	TransformerID	21.07	25.69	29.76	36.66	54.25	55.26
	MMSTGED	15.93	26.68	25.07	39.6	58.13	68.7
	GREEN	21.52	31.96	32.83	36.39	58.67	59.15
	PeriodicMove	21.19	57.11	43.47	53.16	146.53	111.76
	TrajBERT	27.33	43.9	40.72	240.31	79.56	181.36
	TERI	27.33	43.9	40.72	40.46	78.85	69.54
	AttnMove	24.01	98.441	54.55	45.46	89.96	105.72
	Ours	14.55	24.43	22.51	28.24	50.01	50.57

chunks that are easier to process and learn from; without it, the model must learn from the continuous map, leading to substantial degradation in both classification metrics and spatial fidelity. Removing the multiscale module also causes notable loss, especially on spatial discrepancy metrics (MAE, FD), due to the effectiveness of the multiscale module in decomposing, extracting and effectively integrating relevant information from different spatial scales. Similarly, dropping neighbor sampling (w/o neighbor) reduces Acc/F1 and increases MAE/FD, indicating that cross-scale fusion alone is insufficient—explicitly attending to local neighborhoods around start/end points provides crucial fine-grained context. Finally, replacing deformable convolutions with standard convolutions (w/o

deformable) yields a more moderate but consistent performance drop. This suggests that while the grid map and multiscale hierarchy account for most of the gains, offset-aware sampling further refines spatial alignment and reduces trajectory distortion.

Table 3: Ablation Study Results on Geolife Dataset

Model	Acc.	Prec.	Rec.	F1	MAE	FD
Full model	43.57	31.70	31.57	29.49	12.07	20.82
w/o grid map	32.07	19.90	19.97	18.30	20.93	40.33
w/o neighbor	39.67	28.02	28.04	25.69	13.92	26.19
w/o multi scale	39.56	27.93	28.04	25.52	13.90	25.43
w/o deformable	40.87	28.75	28.94	26.36	13.31	23.92

5 Conclusion

In this paper, we address the problem of trajectory recovery in road-network-free, data-scarce environments with *sparse, low-frequency* GPS observations by leveraging web crowdsourced mobility data. We formulate a cross-region trajectory recovery task that emphasizes both point-wise localization and global trajectory fidelity under severe sampling gaps, and propose **RHTR**, a Region-aware Hierarchical Trajectory Recovery model instantiated via a Hierarchical Multi-Scale Grid Transformer. By combining a deformable multi-scale grid encoder, GPS-grid fusion, hierarchical cross-attention, and fine-scale offset regression, **RHTR** explicitly models spatial structure across multiple scales while remaining agnostic to any pre-defined road network and robust to sparsity. Extensive experiments on two real-world datasets (*Geolife* and *Porto*), across multiple sampling regimes and enlarged regions of interest, demonstrate that **RHTR** consistently achieves state-of-the-art performance on both classification-style (Accuracy, Precision, Recall, F_1) and spatial metrics (MAE, FD), with gains most pronounced in the sparsest settings and on the more challenging *Porto* dataset, highlighting the benefits of multi-scale spatial fusion and offset-aware alignment under severe observation gaps.

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